

Footprinting: The Recon Discipline

What footprinting is

The systematic collection of information about a target **before touching it**. Passive footprinting gathers what already exists publicly; active footprinting sends traffic that reveals responses.

	Passive	Active
Contact with target	None (except public records)	Yes (packets you send)
Examples	DNS, WHOIS, search engines, job posts, breach data, cert logs	Port scans, banner grabs, HTTP probes
Legal risk	Low	Higher; needs scope
Detection risk	Low	High (you will be logged)

Objectives

Inventory: domains, subdomains, IPs, ASN, hosting, technologies

People: employees, emails, job titles, personal accounts

Org posture: policies, suppliers, tech stack from job posts

Seams: forgotten subdomains, dev servers, exposed repos

The footprinting loop

```
Domain -> whois (registrar, contacts)
        -> DNS (A, AAAA, MX, TXT/SPF, NS, SOA)
        -> cert transparency (crt.sh) (subdomains you'd miss)
        -> ASN & IP ranges (bgp.he.net, whois)
        -> hosting (AS, cloud provider buckets)
        -> web: robots.txt, sitemap, headers, tech fingerprint
```

Technology fingerprinting

HTTP headers reveal server, framework, versions

`whatweb`, `wappalyzer` (browser), response patterns (WordPress, Shopify)

Error pages, favicon hashes, cookie names – fingerprinting fuel

What good footprinting produces

A **target map** you shall use all engagement: hosts, services, versions, tech, people. The better the map, the smaller the later scan surface. "Footprint first, scan second" is a professional habit, not a formality.

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